🔑 **Answer Key — Teacher Copy · Lesson 3-3**

MSTAR PRACTICE WORKSHEET

# Lesson 3-3: Determine Equivalent Ratios Using Tables

Grade 6 Mathematics · Standard 6.AT.3a

*These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.*

## Part A — Skills Check

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: B

A recipe uses 3 eggs for every 5 cups of flour. How many eggs are needed for 15 cups of flour?

A. 6

B. 9

C. 10

D. 12

The scale factor is 15 ÷ 5 = 3. Multiply the eggs by 3: 3 × 3 = 9 eggs.

**Question 2** SELECTED RESPONSE — CORRECT: A

Which row does NOT belong in a ratio table for 4:5?

A. 6:10

B. 8:10

C. 12:15

D. 16:20

6:10 simplifies to 3:5, not 4:5. The other ratios (8:10, 12:15, 16:20) all simplify to 4:5.

**Question 3** SELECTED RESPONSE — CORRECT: B

A ratio table shows 3:7, 6:14, and 9:21. What is the next row?

A 10:24

B 12:28

C 12:24

D 15:28

The pattern adds 3 to the first value and 7 to the second each time. After 9:21: 9+3=12 and 21+7=28. The next row is 12:28.

**Question 4** SELECTED RESPONSE — CORRECT: B

A ratio table for 2:5 starts with 2:5, 4:10. What is the third row?

A 5:10

B 6:15

C 6:12

D 8:15

The additive pattern adds 2 to the first value and 5 to the second. 4+2=6 and 10+5=15. The third row is 6:15.

**Question 5** SELECTED RESPONSE — CORRECT: B

In a ratio table, the ratio of sugar to butter is 3:2. If you use 12 cups of sugar, how many cups of butter do you need?

A 6

B 8

C 9

D 10

Scale factor: 12 ÷ 3 = 4. Multiply butter by 4: 2 × 4 = 8 cups of butter.

**Question 6** SELECTED RESPONSE — CORRECT: B

A ratio table shows 5:8, 10:16, 15:24. What scale factor was used from the first row to the third row?

A. 2

B. 3

C. 5

D. 8

From 5 to 15: 15 ÷ 5 = 3. From 8 to 24: 24 ÷ 8 = 3. The scale factor is 3.

## Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

**Question 7** TWO-PART QUESTION

**Part A — correct answer: B**

A ratio table pairs 3 cups of flour with 2 cups of water. If the recipe is scaled to use 12 cups of flour, how much water is needed?

A. 6 cups

B. 8 cups

C. 11 cups

D. 24 cups

12 ÷ 3 = 4, so the scale factor is 4. Both terms scale: 2 × 4 = 8 cups of water.

- **A:** That adds 4 to the water instead of scaling it. Flour was MULTIPLIED by 4, so water is too.
- **C:** That adds the same 9 that was added to the flour. Ratio tables scale by multiplying, not adding.
- **D:** That scales the flour term. The scale factor is 12 divided by 3, applied to the 2 cups of water.

**Part B — correct answer: C**

Why is 11 cups an incorrect answer for Part A?

A. It divides instead of multiplying.

B. It uses the wrong conversion factor.

C. It adds 9 to both terms instead of multiplying both by the same factor.

D. It reverses flour and water.

Adding the same amount to both terms changes the relationship. Equivalent ratios require multiplying or dividing BOTH terms by the same non-zero factor.

**Question 8** SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about ratio tables and equivalent ratios:

- **A** Multiplying both terms of a ratio by 3 produces an equivalent ratio.
- **B** Dividing both terms by a common factor produces an equivalent ratio.
- **C** Adding 5 to both terms produces an equivalent ratio.
- **D** Every row of a correct ratio table simplifies to the same ratio.
- **E** A ratio table can be extended in either direction, up or down.

A, B, D, and E are true. C is false — equivalence comes from multiplying or dividing both terms, never from adding.

## Part C — Show Your Thinking

Model answer and scoring guidance.

**Question 9** WRITTEN RESPONSE · 2 POINTS

A recipe uses 4 cups of flour for every 3 cups of sugar. A student filled in a ratio table: 4:3, 8:6, 12:9, 14:12. Find and fix the error in the last row. Explain what went wrong.

**Model answer:** The last row is wrong. The pattern multiplies by the same scale factor. 4×1=4, 4×2=8, 4×3=12, so the 4th row should be 4×4=16 cups of flour and 3×4=12 cups of sugar → 16:12. The student added 2 to 12 instead of adding 4. Or: 12+4=16, not 12+2=14.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

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Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.